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1 %% function f_get_TS_meas_of_target
2 function [TS_meas_target,TS_meas_target_mean] =✓
f_get_TS_meas_of_target(...
3     fEcho_binned, TS_binned, Range, spherePos, doTARGOFFSET,✓
TARGSAMPOFFSET, NCELLBUFF)
4
5 doINDEXWITHTX = false;
6 %%
7 if(doINDEXWITHTX)
8     spherePosINDEX = spherePos.indexWithTx;
9     fprintf('\n[!!!]\nf_get_TS_meas_of_target: using index✓
(including Tx samples).\n');
10 else
11     fprintf('\n[!!!]\nf_get_TS_meas_of_target: using index (not✓
including Tx samples).\n');
12     spherePosINDEX = spherePos.index; % index from PulseCompression
13 end
14
15 %%
16 if(isempty(doTARGOFFSET))
17     doTARGOFFSET=false;
18 end
19 if(isempty(NCELLBUFF))
20     NCELLBUFF = 1;
21 end
22
23 fprintf('\n');
24
25 %%
26 TS_meas_target_mean = zeros(size(fEcho_binned,1));
27
28 %% loop through binned frequencies
29 for( indf = 1 : length(fEcho_binned))
30
31     if(isstruct(TS_binned))
32         sfields = fieldnames(TS_binned); % get fields of TS✓
struct
33         myfield = sfields{indf}; % current field
34         TS_meas_fband = TS_binned.(myfield); % TS_meas samples✓
for this f-band
35     else

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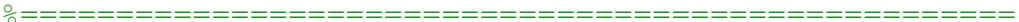
36     TS_meas_fband = TS_binned;
37     end
38     if(doTARGOFFSET) % adjust for sphere pos from 90fm✓
pulse compressed range
39         for(inds = 1 : size( TS_meas_fband,1 ))
40             targindx = round( spherePosINDEX(inds) ) +✓
TARGSAMPOFFSET;
41             COLS_TARG = targindx-NCCELLBUFF : targindx+NCCELLBUFF;
42             %TS_meas_band_itarget(inds, 1:length(COLS_TARG) ) =✓
TS_meas_band( : , COLS_TARG );
43             TS_meas_band_itarget(inds, 1:length(COLS_TARG) ) =✓
TS_meas_fband( inds , COLS_TARG );
44         end
45     else
46         for(inds = 1 : size( TS_meas_fband,1 ))
47             targindx = round( spherePosINDEX(inds) );
48             %TS_meas_band_itarget(inds, 1:length(COLS_TARG) ) =✓
TS_meas_band( : , COLS_TARG );
49             TS_meas_band_itarget(inds, 1:length(COLS_TARG) ) =✓
TS_meas_fband( inds , COLS_TARG );
50         end
51     end
52     if(isstruct(TS_binned))
53         TS_meas_target.(myfield) = TS_meas_band_itarget;
54     else
55         TS_meas_target = TS_meas_band_itarget;
56     end
57
58     if(isstruct(TS_binned))
59         TS_meas_target_q = quantile( TS_meas_target.(myfield),✓
[0.95, 0.99, 0.999]);
60     else
61         TS_meas_target_q = quantile( TS_meas_target, [0.95, 0.99,✓
0.999]);
62     end
63     %%
64     %TS_meas_target_mean(indf) = 10*log10( mean(10.^(reshape✓
(TS_meas_target_q, [],1)/10)));
65     %%
66     % Sven, 2020-01-08:
67     % When we calculate the return signal from the reference target✓

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sphere,
68     % the return signal typically is distributed over three cells.
69     % I think we want to total returned energy when estimate the
target strength
70     % and hence I suggest to modify line 63 in
f_get_TS_meas_of_target.m
71     % to this line: TS_meas_target_mean(indf) = 10*log10(mean(sum
(10.^((TS_meas_target_q')/10))));
72     %The target strength should increase when we consider all parts
of
73     % the return signal, so the signal from the three cells/bins
should
74     % be added (correctly in linear space) before we take the
average.
75     % This will change the calibration values significantly.
76     TS_meas_target_mean(indf) = 10*log10(mean(sum(10.^
((TS_meas_target_q')/10))));
77
78     fprintf('(findex %d) TS_meas_target_mean = %.2f.\n', indf,
TS_meas_target_mean(indf));
79
80     %% Figure =====
81     matnames = {'TS_meas_band'} ;
82     f_plot_echogram_maxamp( TS_meas_fband , Range , matnames);
83     %
84     % plot with NO OFFSET
85     hold on;
86     plot( mean(TS_meas_fband(:,mode( spherePosINDEX ))), mean(Range
(mode( spherePosINDEX ))),...
87         'om','MarkerSize',10,'LineWidth',0.8);
88     plot( mean(TS_meas_fband(:,mode( spherePosINDEX ))), mean(Range
(mode( spherePosINDEX ))),...
89         'xm','MarkerSize',10,'LineWidth',0.8);
90
91     % plot with TARGETOFFSET
92     hold on;
93     plot( TS_meas_target_mean(indf) , mean(Range(mode(
spherePosINDEX )+TARGSAMPOFFSET)),...
94         'og','MarkerSize',10,'LineWidth',1.2);
95
96     newfigname=[get(gcf,'Name') ' (f_get_TS_meas)'];

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97     set(gcf, 'Name', newfigname);  
98     %=====  =====  
99 end  
100 fprintf('\n');  
101 end
```